House Price Prediction Using Machine Learning: A Comparative Study of Regression Models on the Ames Housing Dataset

Shravan pawar
Mathematics and Computing
Indian Institute of Technology,Dharwad
Dharwad, India
shravan050407@gmail.com

***Abstract*—House price estimation is an important regression problem that involves understanding the relationship between property characteristics and their market values. This project develops a machine learning pipeline to predict residential house sale prices using the Ames Housing dataset. The dataset contains numerical and categorical information related to property size, quality, location, construction, garages, basements, and other facilities.**

The project includes exploratory data analysis, missing-value treatment, categorical encoding, model training, hyperparameter tuning, and performance evaluation. Linear Regression, log-target Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting were evaluated. Pairwise feature interactions and unusual observations were also investigated.

The final selected model is a Linear Regression model trained on the logarithmically transformed sale price. After excluding two unusual observations from the training data, the model achieved a validation Mean Absolute Error (MAE) of $15,214.19, Root Mean Squared Error (RMSE) of $23,150.51, and coefficient of determination (R^2) of 0.9301. Five-fold cross-validation on the cleaned training data produced a mean MAE of $14,201.71 and mean R^2 of 0.9260. The trained model was saved using Joblib and used to generate 1,459 predictions for the Kaggle test dataset.

***Index Terms*—House Price Prediction, Machine Learning, Regression, Ames Housing Dataset, Linear Regression, Random Forest, Gradient Boosting, Feature Engineering, Cross-Validation.**

I. Introduction

Real estate price estimation is a challenging problem because the value of a residential property depends on several factors. These factors include living area, construction quality, location, property age, number of rooms, garage facilities, basement area, and other structural characteristics.

Traditional estimation methods may depend heavily on manual assessment and local market knowledge. Machine learning provides an alternative approach by learning relationships between historical property characteristics and their observed sale prices.

The objective of this project is to develop a regression-based machine learning system that predicts the sale price of houses using the Ames Housing dataset.

The project investigates multiple regression algorithms and preprocessing techniques to identify a suitable predictive model. It also studies prediction errors, unusual observations, and the effect of transforming the target variable.

II. Problem Statement

The objective is to predict the sale price of a residential property based on its available characteristics.

Let the input feature vector be represented as:

$$X = [x_1, x_2, x_3, \ldots, x_n] \tag{1}$$

where the features represent property characteristics such as overall quality, living area, construction year, garage capacity, and neighborhood.

The regression model learns a function:

$$\hat{y} = f(X) \tag{2}$$

where $\hat{y}$ represents the predicted sale price.

The project aims to:

- Understand and preprocess the Ames Housing dataset.
- Handle missing numerical and categorical values.
- Train and compare different regression algorithms.
- Evaluate prediction performance using MAE, RMSE, and R^2.
- Investigate prediction errors and unusual observations.
- Select and save a suitable final predictive model.
- Generate predictions for unseen Kaggle test records.

III. Dataset Description

The project uses the Ames Housing dataset, which contains residential property information from Ames, Iowa.

The training dataset contains 1,460 records and 81 columns, including the target variable `SalePrice`. The Kaggle test dataset contains 1,459 records and 80 columns, without the target variable.

The dataset includes numerical and categorical attributes.

A. Important Features

Some important numerical features include:

- `OverallQual`: Overall material and finish quality.
- `GrLivArea`: Above-ground living area.
- `GarageCars`: Garage capacity in number of cars.
- `GarageArea`: Garage area in square feet.
- `TotalBsmtSF`: Total basement area.
- `YearBuilt`: Original construction year.
- `LotArea`: Total lot area.

Categorical features include neighborhood, building type, exterior quality, basement quality, garage type, and sale condition.

The target variable is:

$$y = \texttt{SalePrice} \tag{3}$$

IV. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of sale prices, identify missing values, and examine relationships between property characteristics and the target variable.

A. Sale Price Distribution

The training dataset has the following target statistics:

TABLE I
SALE PRICE STATISTICS

Statistic	Value (USD)
Count	1460
Mean	180,921.20
Standard Deviation	79,442.50
Minimum	34,900
Median	163,000
Maximum	755,000

The original sale price distribution has a skewness of approximately 1.88, indicating a positively skewed distribution.

A logarithmic transformation was investigated to reduce this skewness.

B. Correlation Analysis

Correlation analysis was performed to identify numerical features associated with sale prices.

The strongest observed correlations include:

TABLE II
IMPORTANT NUMERICAL FEATURE CORRELATIONS

Feature	Correlation
OverallQual	0.7910
GrLivArea	0.7086
GarageCars	0.6404
GarageArea	0.6234
TotalBsmtSF	0.6136
1stFlrSF	0.6059
FullBath	0.5607
TotRmsAbvGrd	0.5337
YearBuilt	0.5229
YearRemodAdd	0.5071

Overall quality and above-ground living area showed strong positive correlations with sale prices.

Correlation indicates association and does not by itself establish causation.

V. Data Preprocessing

Data preprocessing was performed using Scikit-learn pipelines and a ColumnTransformer.

The identifier column `Id` and target-derived columns were excluded from the predictor set.

A total of 79 predictor features were used:

- 36 numerical features.
- 15 categorical features representing the absence of a property feature.
- 28 other categorical features.

A. Missing Value Treatment

Missing values were analyzed before applying imputation.

Some missing categorical values represent the absence of a property feature rather than unavailable information. For example, a missing pool quality value can indicate that the property does not have a pool.

The following strategies were used:

- Numerical features: Median imputation.
- Absence-related categorical features: Constant imputation using the value “None”.
- Other categorical features: Most-frequent imputation.

B. Categorical Encoding

One-Hot Encoding was applied to categorical variables.

This converts each category into a separate binary feature, avoiding the artificial numerical ordering that can occur when categories are represented using arbitrary integers.

The encoder used:

```
OneHotEncoder(handle_unknown="ignore")
```

This setting allows the model to process categories that were not encountered during training.

C. Preprocessing Pipeline

A ColumnTransformer combines the numerical and categorical preprocessing pipelines.

The preprocessing operations and regression model were combined into a Scikit-learn Pipeline. This ensures that the preprocessing steps are fitted on training data and applied consistently to validation and test data.

The dataset was divided using an 80:20 split:

- Training records: 1,168
- Validation records: 292
- Random state: 42

VI. Target Variable Transformation

The original sale price distribution was positively skewed. To investigate whether a more symmetric target distribution would improve model performance, a logarithmic transformation was applied.

The transformation is:

$$y_{\log} = \log(1 + y) \tag{4}$$

Here, $\log(1+y)$ is implemented using NumPy's `log1p()` function.

The inverse transformation is:

$$\hat{y} = \exp(\hat{y}_{\log}) - 1 \tag{5}$$

This is implemented using `expm1()`.

The transformation reduced the observed target skewness from approximately 1.88 to approximately 0.12.

The logarithmic target was evaluated by converting predictions back to the original sale price scale.

VII. Model Development and Comparison

Multiple regression algorithms were implemented and evaluated using the same training and validation split.

A. Linear Regression

Linear Regression was used as the baseline model.

It estimates the relationship between input features and the target variable using:

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i \tag{6}$$

The baseline model was trained directly on the original sale price values.

B. Log-Target Linear Regression

A second Linear Regression model was trained using log-transformed sale prices.

The predictions were converted back to the original price scale before evaluation.

This model improved the validation results compared with the baseline Linear Regression model.

C. Ridge Regression

Ridge Regression was tested to investigate the effect of regularization.

It minimizes the residual sum of squares with an additional penalty on the coefficient magnitudes:

$$J(\beta) = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{n} \beta_j^2 \tag{7}$$

GridSearchCV was used to select the regularization parameter from the values 0.1, 1, 10, and 100.

The best value was $\alpha = 100$.

D. Random Forest Regression

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees.

The initial model used 300 trees and was trained on log-transformed sale prices.

RandomizedSearchCV was then used to investigate:

- Maximum tree depth.
- Minimum samples required for a split.
- Minimum samples per leaf.
- Maximum number of features considered for splits.

The best configuration obtained was:

```
max_depth = 10
min_samples_split = 2
min_samples_leaf = 2
max_features = 0.7
```

E. Gradient Boosting Regression

Gradient Boosting was also evaluated.

The model builds an ensemble of decision trees sequentially. Each subsequent tree attempts to correct errors made by the existing ensemble.

The configuration used 300 estimators, a learning rate of 0.03, maximum depth of 2, and minimum leaf size of 5.

F. Pairwise Feature Interactions

PolynomialFeatures was used to investigate pairwise interactions between features.

The configuration was:

```
degree = 2
interaction_only = True
include_bias = False
```

The interaction model was trained using log-transformed sale prices. Its performance was compared with the log-target Linear Regression model.

VIII. Model Evaluation and Results

The models were evaluated using three standard regression metrics.

A. Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{8}$$

B. Root Mean Squared Error

Root Mean Squared Error penalizes larger errors more strongly because the errors are squared before averaging.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{9}$$

C. Coefficient of Determination

The coefficient of determination measures the proportion of target variance explained by the model relative to a mean-prediction baseline.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{10}$$

D. Model Comparison

Table III presents the validation performance of the evaluated models. All metrics are reported on the original sale price scale.

The log-target Linear Regression model achieved the best validation performance among the initial model comparisons, with an MAE of \$16,969.82 and an R^2 of 0.9155.

The tuned Random Forest model improved over the default Random Forest configuration, but did not outperform the log-target Linear Regression model on validation MAE.

The pairwise interaction experiment also did not improve the validation MAE.

IX. Feature Importance and Error Analysis

Prediction errors were analyzed to understand where the model performed well and where it struggled.

A. Random Forest Feature Importance

Feature importance scores from the tuned Random Forest model were examined.

The most important features included:

TABLE IV
Top Random Forest Feature Importances

Feature	Importance
OverallQual	0.4146
GrLivArea	0.1556
YearBuilt	0.0597
GarageCars	0.0423
TotalBsmtSF	0.0378
GarageArea	0.0305
1stFlrSF	0.0227
BsmtFinSF1	0.0209

Overall quality and above-ground living area were the most influential features according to the fitted Random Forest model.

These importance scores describe the model's use of features and should not be interpreted as causal effects.

B. Prediction Error Analysis

For the log-target Linear Regression model, the validation error was calculated as:

$$e_i = y_i - \hat{y}_i \tag{11}$$

A positive residual indicates underprediction, while a negative residual indicates overprediction.

The model achieved:

- Mean Absolute Error: \$16,969.82
- Root Mean Squared Error: \$25,451.67
- Mean Absolute Percentage Error: 10.32%

The error analysis showed that predictions for expensive houses were generally more difficult. The validation group containing houses above \$400,000 had a mean absolute error of approximately \$76,252.89, although this group contained only seven houses.

C. Residual Analysis

Residual plots were examined against:

- Above-ground living area.
- Overall house quality.
- Actual sale price.

The analysis showed that prediction errors were not uniform across all property groups. Houses with very high overall quality had larger average absolute errors in some groups, although the number of observations in those groups was small.

X. Investigation of Unusual Observations

Out-of-fold predictions were generated using five-fold cross-validation to investigate observations with particularly large prediction errors.

Two training observations, with indices 1298 and 523, were identified for further inspection.

These records had unusually large living areas and basement areas compared with typical properties in the dataset, but their recorded sale prices were relatively low compared with the model predictions.

The observations were investigated using their property characteristics and feature distributions.

Since unusual observations can represent either valid rare properties or potential data anomalies, they should not automatically be considered incorrect.

For an experimental comparison, these two observations were excluded from the training data only. The validation set was kept unchanged.

The cleaned training set contained 1,166 records, while the validation set continued to contain 292 records.

The resulting model achieved:

- MAE: \$15,214.19
- RMSE: \$23,150.51
- R^2: 0.9301

Compared with the original log-target Linear Regression model, the cleaned-data experiment improved validation MAE and RMSE.

However, this improvement does not establish that removing unusual observations is universally beneficial. The final model's performance may be less representative of rare properties similar to the excluded records.

XI. Cross-Validation

Five-fold cross-validation was used to assess model performance across multiple training and validation partitions.

TABLE III
COMPARISON OF REGRESSION MODELS

Model	MAE (USD)	RMSE (USD)	R^2
Linear Regression	20,472.74	31,045.85	0.8743
Log-Target Linear Regression	16,969.82	25,451.67	0.9155
Ridge Regression	17,822.58	29,421.19	0.8871
Random Forest	17,504.93	30,015.38	0.8825
Tuned Random Forest	16,992.97	28,980.76	0.8905
Gradient Boosting	18,112.12	31,851.38	0.8677
Pairwise Interaction Linear Regression	17,961.23	29,253.01	0.8884

The training data was divided into five folds. In each iteration, four folds were used for training and the remaining fold was used for validation.

The process was repeated until every fold had served as the validation fold.

Cross-validation was performed on both the original and cleaned training datasets using the log-target Linear Regression pipeline.

A. Cross-Validation Results

TABLE V
FIVE-FOLD CROSS-VALIDATION RESULTS

Metric	Original	Cleaned
Mean MAE (USD)	18,172.86	14,201.71
MAE Std. Dev.	1,134.76	540.62
Mean RMSE (USD)	42,723.27	20,960.95
RMSE Std. Dev.	22,872.31	1,128.87
Mean R^2	0.6303	0.9260

The cleaned-data cross-validation results showed lower average errors and less variation across folds.

These results are conditional on excluding the two identified observations from the training dataset. They should not be interpreted as evidence that the model will perform equally well on every possible house or on future data containing similar unusual cases.

XII. FINAL MODEL AND IMPLEMENTATION

Based on the experiments, log-target Linear Regression trained on the cleaned training dataset was selected as the final model.

The implementation uses a Scikit-learn TransformedTargetRegressor with a preprocessing pipeline and Linear Regression estimator.

The target transformation and inverse transformation are handled by the estimator:

```
TransformedTargetRegressor(
    regressor=Pipeline([
        ("preprocessor", preprocessor),
        ("model", LinearRegression())
    ]),
    func=np.log1p,
    inverse_func=np.expm1
)
```

The model was fitted using the cleaned training data.

A. Final Model Performance

TABLE VI
FINAL MODEL PERFORMANCE

Metric	Value
Training MAE	\$11,313.07
Validation MAE	\$15,214.19
Validation RMSE	\$23,150.51
Validation R^2	0.9301

The final model explains approximately 93.01% of the variation in the validation-set sale prices relative to a mean-prediction baseline.

This value should not be interpreted as 93.01% prediction accuracy.

B. Model Serialization

The trained model was saved using the Joblib library in the following file:

`ames_housing_final_model.pkl`

Saving the complete estimator preserves the fitted preprocessing steps and regression model together.

XIII. PREDICTION ON THE KAGGLE TEST DATASET

The saved final model was applied to the Kaggle test dataset, which contains 1,459 records.

Since the test dataset does not contain the target variable, actual prediction errors could not be calculated for these records within the notebook.

The model generated 1,459 sale price predictions.

The prediction range was approximately:

- Minimum predicted price: \$53,431.98
- Maximum predicted price: \$1,800,207.51

A submission file was generated with two columns: `Id` and `SalePrice`.

The output file was saved as:

`submission.csv`

The notebook confirms that the submission file was created successfully. No Kaggle leaderboard score is reported in this project.

XIV. Limitations

Although the final model achieved promising validation and cross-validation results, several limitations should be considered.

- The model was developed using a single historical housing dataset from Ames, Iowa. Its performance may not generalize to other geographical locations or market conditions.
- The validation set was used during iterative model comparison and experimentation. Therefore, its final performance is not an independent, untouched test estimate.
- Two unusual observations were excluded from the final training data. These may represent valid rare properties rather than incorrect records.
- Linear Regression assumes a linear relationship in the transformed target space and may not capture every complex relationship between property features.
- The Kaggle test predictions were generated, but no ground-truth labels were available in the notebook to evaluate their actual prediction errors.
- Feature importance values from Random Forest describe model-specific importance and do not establish causation.

XV. Conclusion

This project developed a complete machine learning workflow for residential house price prediction using the Ames Housing dataset.

The workflow included dataset exploration, missing-value analysis, numerical and categorical preprocessing, target transformation, model training, hyperparameter tuning, error analysis, outlier investigation, and cross-validation.

Several regression algorithms were compared, including Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting.

The experiments demonstrated that applying a logarithmic transformation to the target variable improved the validation performance of Linear Regression.

The final selected model was log-target Linear Regression trained after excluding two unusual observations from the training data. It achieved a validation MAE of \$15,214.19, RMSE of \$23,150.51, and R^2 of 0.9301.

Five-fold cross-validation on the cleaned training data also showed promising results, with a mean MAE of \$14,201.71 and mean R^2 of 0.9260.

Finally, the trained model was serialized using Joblib and used to generate predictions for all 1,459 records in the Kaggle test dataset.

The project demonstrates how preprocessing, target transformation, model comparison, and careful error analysis can contribute to developing a practical regression-based prediction system.

References

[1] D. De Cock, "Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project," *Journal of Statistics Education*, vol. 19, no. 3, 2011.

[2] Kaggle, "House Prices: Advanced Regression Techniques," Kaggle Competition. [Online]. Available: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[4] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.